

A Comparative Study of Simulated Augmented Reality Displays for Vehicle Navigation

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ABSTRACT

In this paper we report on a user study in a simulated environment that compares three types of Augmented Reality (AR) displays for assisting with car navigation: Heads Up Display (HUD), Head Mounted Display (HMD) and Heads Down Display (HDD). The virtual cues shown on each of the interface were the same, but there was a significant difference in driver behaviour and preference between interfaces. Overall, users performed better and preferred the HUD over the HDD, and the HMD was ranked lowest. These results have implications for people wanting to use AR cues for car navigation.

Author Keywords

Augmented Reality; Car Navigation, Head Up Display.

ACM Classification Keywords

H.5.2 User Interfaces; H.5.1 Multimedia Information Systems - Artificial, augmented, and virtual realities.

INTRODUCTION

This paper reports on research to compare different Augmented Reality (AR) displays for vehicle navigation. AR technology is usually considered to be a live view of the real world, onto which virtual information is superimposed, and can be interacted with (Kipper & Rampolla, 2012). One possible use of AR in cars is to provide navigation cues, such as virtual arrows (Tonniss et al., 2007), or lines on the road (Hu & Uchimura, 2004).

There are a number of ways to implement an in-car AR navigation interface. One of the most common is by using a Head Down Display (HDD) mounted on the dashboard showing an AR enhanced live video view (see Figure 1 top). This display shows AR cues on a live camera view of the real world. A common example of this is the use of in-car displays for parallel parking by using AR imagery with reverse cameras. The main advantage is that the display will not block the driver's view of the real world. However the driver does need to take his or her eyes off the road to view the display.

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Figure 1. Simulated AR views (HDD, HUD, and HMD from top to bottom).

Another approach is to project information onto a Heads Up Display (HUD) on the windscreen in the driver's line of sight (see Figure 1 middle). This reduces the time the driver needs to take their eyes off the road and provides a large field of view. However, there are a few challenges, such as the level of visibility in direct sunlight, and the obstruction of real world elements with the AR cues.

Finally, the driver could wear a lightweight Head Mounted Display (HMD), such as Google Glass (<http://glass.google.com>), to create a personal virtual display overlaying the real world (see Figure 1 bottom). The main benefit of using a HMD is that the user can see navigation information regardless of where they are looking. However, the display may block or at least cover certain portion of the driver's real world view.

Since there are a number of different ways to show AR navigation cues in a car, it is important to research the

effect on driver behaviour of each of these options. Some of the existing research in using AR for in-car navigation compares using a HUD with an in-car HDD (Charissis et al., 2008; Liu & Wen, 2004). There is also research comparing using AR cues instead of map navigation (Yamaguchi et al., 2007). However, to the best of our knowledge, we are unaware of previous research that directly compares the same AR interface on a HMD, HUD and HDD for in-car navigation. Thus, the main contribution of this paper is to provide comparisons of HMD, HUD and HDD display technologies for AR navigation.

Testing of AR navigation interfaces in real world driving conditions is a challenge due to safety concerns, technical difficulties of producing bright AR images, and having consistent environmental conditions for user studies. Thus, like other previous research (e.g., Chu & Joseph, 2008; Kim & Dey, 2009; Medenica et al., 2011; Ng-Thow-Hing et al., 2013), we conduct our user studies in an immersive driving simulation. In the future we aim to conduct user studies in real driving conditions outdoors.

RELATED WORK

Using AR for car navigation is an area that has been well studied. This section reviews previous related research in using AR interfaces for vehicle navigation, including earlier user experiments and evaluation methods.

Using AR Cues for Car Navigation

In a vehicle, the user's tasks fall into primary, secondary and tertiary categories (Geiser, 1985). The primary task involves the manoeuvring of the vehicle while keeping an eye on the surroundings and distance to other cars and objects. Secondary tasks are those that supplement the primary tasks like controlling the turning indicators or windshield wipers. Finally, tertiary tasks do not have any relationship with driving, such as using the car radio. Each of these tasks typically use different physical areas in the driver's space for information display and input. The primary task uses the windshield, while secondary tasks use the in-car display and controls near the windshield, and tertiary tasks use other controls. HUD and HMD allow information to be placed in the primary task area, while HDD place information on a physical display in the secondary task area inside the car.

The first vehicle HUDs were developed in late 80's with the Oldsmobile Cutlass Supreme (Horrey et al., 2003) where techniques adopted from the aviation industry were used to project information onto the car windshield. The first use of HMDs in cars dated back to the late 1990s with Regenbrecht's AR Training System for driver safety (Regenbrecht et al., 2005). From the early 2000's, HDD based AR systems such as VICNAS (Hu & Uchimura, 2004) were developed that added AR navigation cues onto live video shown on in-car console display.

Using HDD for AR Navigation

One of the easiest ways to provide AR navigation cues is to add virtual information to a live camera view shown in an in-car display. For example Narzt et al. (2003) developed the INSTAR system that used an AR overlay on a live video feed shown on a small display mounted on

the car dashboard. This system combined car GPS and orientation sensors with points of interest and map information from a spatial database for navigation. Around this time researchers were also exploring using AR overlay on a HDD to have virtual cues that aligned with the real world (Siemens, 2002), rather than just using screen aligned navigation cues. For example, the VICNAS system (Hu & Uchimura 2004) used hybrid tracking with GPS, inertial sensing and computer vision combined together to align virtual navigation arrows with the real terrain. In the AR-Navi HDD system (Yamaguchi et al., 2007), researchers extended this by having both an ego-centric AR view and a top-down map view.

Recently, commercial uses of HDD for in-car navigation have become common. One of the earliest uses of AR in production cars has been the use of reversing cameras for parking assistance. For example, Ford (2008) implemented this in their Fiesta 2011 models with forward and reverse parking assistance. There are also applications available that can turn any smart phone into an AR navigation aid. The 2010 Wikitude Drive application, (<http://www.wikitude.com/wikitude-drive>) provides turn-by-turn navigation using AR cues on a user's smart phone attached to the car dashboard.

Using HUD for AR Navigation

There has been a lot of research on using the windshield as a HUD for vehicles. A HUD provides a wider field of view than the HDD, overlays graphics on a direct view of the real world, and is located in the driver's primary task space, and so doesn't require the driver to take their eyes off the road. For example, the University of Minnesota's driver assistance system uses AR technology, and an optical HUD for collision avoidance (GPS World, 2010). The system showed lane markings to assist the driver with navigation during difficult driving conditions.

Researchers have developed HUD interfaces specifically to provide car navigation cues. For example Tonniss et al. (2007) created an interface that projects AR cues on the windshield that align with the road surface. Narzt et al. (2003) goes beyond this to not only show navigation arrows, but also highlights freeway exits, points of interest such as petrol stations, and alerts to potential hazards. More recently, the DICE AR system showed how the whole windshield of a car could be used to present driver information (Jungwirth, 2012).

Using the windshield as a display has also been explored by commercial brands in recent years. For example, Jaguar's Land Rover launched a concept video showing AR imagery on a windshield in a car (Jaguar, 2014). This display is interactive and has features like virtual racing lines that change colour to indicate optimum braking, and virtual road cones to train novice drivers. Similarly, Pioneer (2012) has launched their Cyber Navi line which features both a HUD on the car's sun visor and a fixed console on the dashboard that captures video from a camera mounted near the rear view mirror. This camera scans the road ahead and provides a live video feed to the console, and uses object recognition to detect street signs and traffic and presents this information with AR cues.

Using HMDs for AR Navigation

There are far fewer examples of AR interfaces using HMDs in vehicles. Ferreira et al. (2013) describes the use of a video see-through HMD to enable a driver to see through vehicles in front of them. In this case the wireless video is sent from the lead vehicle to the one following behind to show the scene ahead. Santos (2014) describes an extension of this where the integrated head tracking in the HMD could assist with knowing where the user was looking and what virtual cues should appear in his or her field of view. Regenbrecht et al. (2005) described how a similar system could be used to provide driver training, in this case using a video see-through HMD to present a number of dangerous driving situations. However none of these systems provided navigation cues to the user.

One of the few AR HMD systems demonstrating an in-car navigation aid is the work of Foxlin et al. (2014). They developed a precise head tracking system for optical see-through HMDs in vehicles, and showed virtual tunnel navigation cues that guide the user in the direction they should go. However the main focus of their work was on the in-vehicle tracking system and navigation interface.

Previous Comparative Studies

There have been many studies that have compared using HUDs to HDDs in driving tasks. These studies typically record the driver performance, the number of errors made, the driver workload, and the user preference. For example, Doshi et al. (2009) conducted a study with the windshield as a display and using AR elements to convey an alert when the driver drove above the speed limit. They found that user's had a faster reaction time using the HUD than when using a HDD. Similarly Liu and Wen (2004) found that truck drivers were able to control their speed better when shown speed information in a HUD rather than a HDD, with a 1.0 second saving in reaction time.

Medenica et al. (2011) used a driving simulator to measure performance with an AR HUD compared to two types of HDD navigation aids. Using eye-tracking technology they found that with the AR HUD drivers were able to keep their eyes on the road significantly longer than with a HDD and experienced significantly less workload. Other experiments include the work of Charissis et al. (2008) that found an AR HUD produced significantly less collisions in low visibility conditions than a HDD, and Kim and Dey (2009) showing fewer navigation and driving errors with a novel AR HUD compared to standard HDD.

Some research has also been conducted on comparing AR to non-AR interfaces for the same display type. For example, Park et al. (2007) compared driver performance when using world-registered arrows to screen-registered 2D icons on the HUD. They found that drivers had a significantly faster response time to lane changing information when shown with the world-registered cues. Similarly, Sawano and Okado (2006) conducted a study comparing an AR HDD to a more traditional HDD with non-AR map interface. Users found the AR HDD interface to be more understandable, although there was no significant difference in driver performance.

In summary, over the past 20 years researchers have developed a range of different AR interfaces for driver assistance, using HDD, HUD and HMD techniques. User studies show the HUDs provide better performance than HDDs, but further research needs to be done with HMDs. It seems that there has been little or no work previously published comparing the same driving AR interface in a HDD, HUD and HMD under the same conditions, either in a driving simulator or in real driving conditions. This is the research gap that our work seeks to address.

PROTOTYPE DESIGN AND IMPLEMENTATION

To test different AR display configuration, a virtual driving simulator was developed based on the Oculus Rift DK1 HMD and Ferrari Thrustmaster GT steering wheel (see Figure 2). The Rift is a wide field of view stereo HMD (110 degree diagonally and 90 degree horizontal) with 640x800 pixels per eye. The device also comes with an orientation and motion sensor (sampling at 1000Hz) to track the rotation of the user's head as they look around the virtual world. For audio we used a noise-cancelling headset (Audio Technica ATH-ANC33iS).



Figure 2. HMD and steering wheel used.

The virtual simulation was developed using the Unity3D v4.5 game engine (<http://unity3d.com>) and the UnityCar 2.2 (<http://unitypackages.net/unitycar/joomla/index.php>) plug-in that provides a realistic vehicle simulation. The car used was a left-hand drive model (see Figure 1). A virtual human was added to the driver seat and the behaviour of the car was connected to the steering wheel and pedals. The car model was also tweaked to add rendered textures with cameras to the side and central rear view mirrors. A forward looking camera was added for the AR view.

The city environment used for the driving simulator was the Unity City terrain from UnityCar 2.2. This is an extensive city model that has various roads and buildings. The map has a set of roads that could be used to navigate to a fixed position using different routes (see Figure 3).

AR Interface

Most of the development work of the prototype involved building three simulated AR displays, a HUD, HDD and a HMD. The implementation of each AR display was done using the NGUI plug-in for Unity (<https://www.assetstore.unity3d.com/en/#!/content/2413>) for implementing 2D user interfaces for the displays.

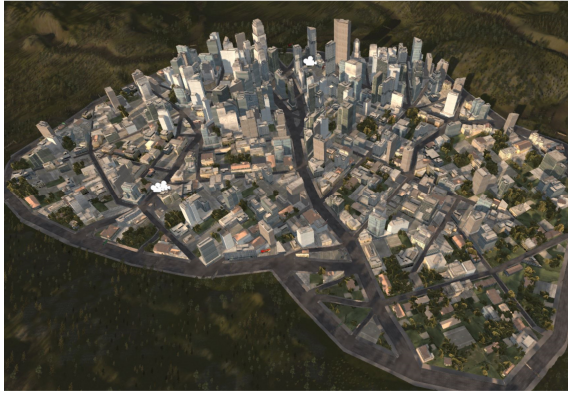


Figure 3. Virtual city in the prototype.

The AR interface was used to display speed, the current gear, and direction information (see Figure 4). Two-dimensional arrows were used to show the turn information. The driver's speed was colour coded to easily identify how close they were to the speed limit (60km/h); Green when below the speed limit, Yellow nearing the speed limit and Red when above the speed limit. Finally, the display area for each AR display was the same size/field of view so that there would be no bias.

Note that while world-registered cues could take greater advantages of AR, we only used screen-registered cues to make the study simpler. We plan to further investigate world-registered cues in the future.



Figure 4. AR interface design.

The Windshield HUD simulation was displayed at the bottom of the windshield (see Figure 1 middle). For the HMD simulation, a display with a 20 degree field of view was shown on the top right corner of the user's view, similar to Google Glass. The display was fixed to the user's viewpoint and so always in view when they turned their head (see Figure 1 bottom). The HDD display had its display screen fixed into the console mounted on the dashboard (see Figure 1 top). This was placed at approximately 30 degrees below the eye level.

USER EXPERIMENT

We conducted a user experiment to compare between the different simulated AR displays for vehicle navigation. The goal was to compare the effects of using different types of AR displays for navigation while driving.

Experiment Design

A within participants experiment was designed with the following three conditions: HUD, HMD, and HDD. In each condition, participants were asked to drive fixed path in the virtual simulation following the AR navigation cues to get to the destination as fast and safe as possible. Participants had to fulfil three tasks:

1. Follow navigational instructions on the display
2. Keep to the speed limit of 60km/h
3. Look around for virtual characters in the scene

Virtual characters were placed on certain positions throughout the city. The participant had to look around for these characters as they drove and click a button on the steering wheel each time they saw them. This was to measure how attentive the user was to his or her surroundings and to ensure that they looked around at intersections and corners similar to real driving conditions.

The user study was conducted in a small room (3m x 5m), where participants sat on a chair using the prototype.

Procedure

The whole procedure took about 40 minutes. The experiment started with the participant reading the information sheet and signing the consent form. The procedure was then explained along with how each AR display looked and the tasks. Then the researcher assisted the participant in putting on the equipment. The participant first drove freely for 2 minutes to familiarize themselves with the system. Then they tried three conditions in order defined with a balanced Latin Square design to counterbalance the order effect. It took about 7 minutes for the participants to complete the task for each condition. The task results were logged into a text file.

After completing the task in each condition, participants were asked to fill a questionnaire on comfort, usability and efficiency of each AR display. At the end of all three test conditions, the participants were asked to fill in a questionnaire with their demographic data including age, gender, experience with driving, and previous experience with the Oculus Rift and Virtual Reality (VR). They were also asked to rank the three conditions based on their preference.

There were several objective measurements that were recorded. One was the number of navigational errors, based on the number of wrong turns the participants made. The number of correct turns that a user made was also recorded. Since the length of the route changed as the user made a wrong turn, we used the ratio of the number of wrong turns to total number of turns made.

As the participant tried navigating through the virtual city, they were expected to maintain a speed of 60km/hr. Every time they exceeded 62km/hr, the application logged the instance. It also recorded the time the user spent above this speed until they went back to below the speed limit. This measured the time the user took to realize that they had crossed the speed limit.

One of the main requirements for a driver was to be aware of their surroundings and identifying hazards. This was tested by placing 20 virtual characters at different positions. The participants were asked to look around as they were driving and to click a button when they spotted the character. The number of clicks was logged to help understand the level of concentration each AR display required from the participant while driving. When a participant made a wrong turn, they were directed to get

back to where they made the wrong turn so that the number of characters to spot was kept as the same.

We collected subjective feedback through questionnaires about the participant's view of each of the condition. The questionnaire included 7-point Likert scale rating questions (answered on a scale from 1 (totally disagree) to 7 (totally agree)), ranking, and open questions for collecting qualitative feedback. See Table 1.

No.	Rating Question Statement
Q1	The surroundings were clearly visible while using the display
Q2	The graphics on the display are clearly visible .
Q3	Using this display distracted me from driving.
Q4	It was easy to navigate using the display.
Q5	This display helps me in multitasking between maintaining ideal speed, looking around, and navigation.
Q6	The user interface of the display was useful in helping me navigate properly.
Q7	I would feel comfortable if asked to wear this AR display on a regular basis when I drive.

Table 1. Likert scale rating questions.

Participants

Three participants were recruited for a pilot test. The results obtained from the pilot test were not included in the results of the study, but were used to test the system and experimental procedure before the main experiment.

In the main experiment, 18 participants (14 male and 4 female) participated in the user study aged between 20 and 35 years (Mean: 25.5 years). Table 2 summarises demographic characteristics of the participants. Their driving experience ranged from 1 to 10 years. Several participants had used the Oculus Rift before, but none had extensive VR experience. Around 33% of participants wore glasses while driving, but they could wear their glasses together with the HMD.

Property	Description
Age	20 - 35 years old (Mean: 25.5)
Gender	14 male (78%), 4 female (22%)
Driving Experience	1 - 10 years (Mean: 6.8)
Eye sight	12 normal, 6 with prescription glasses

Table 2. Demographic characteristics of participants.

EXPERIMENT RESULTS

Overall we found that the HUD produced the least navigational errors, while participants exceeded the speed limit most in the HDD. In terms of subjective feedback, users felt that the HUD scored the best on most of the questions in the questionnaire, and users ranked the HUD as their most preferred and the HMD as the least.

Objective Measures

The objective measures recorded by the system were the ratio of navigational errors made, the number of times exceeding the speed limit, the time spent above the speed limit, and the number of character models detected. Shapiro-Wilk test ($\alpha=.05$) showed all three measures were not normally distributed (note that counting events usually follow Poisson distribution), hence we used a Friedman test ($\alpha=.05$) to analyse the results instead of ANOVA. For post hoc tests we used Wilcoxon signed rank tests with Bonferroni correction ($\alpha=.0167$).

Figure 5 shows the results of the navigational error (number of wrong turns divided by all turns). The medians [IQR] of the navigation error for the HDD, HMD, and HUD conditions were 0.43 [0.09-0.56], 0.41 [0.13-0.72], and 0.10 [0.05~0.30], respectively. A Friedman test found a significant difference between the conditions in navigational error made ($\chi^2(2)=15.746$, $p<.001$). Post hoc analysis with Wilcoxon signed-rank tests with Bonferroni correction showed participants made significantly fewer errors with the HUD condition compared to the HDD ($Z=-2.676$, $p=.007$) and HMD ($Z=-3.027$, $p=.002$). There was no significant difference between the HDD and HMD ($Z=-1.111$, $p=.267$) conditions.

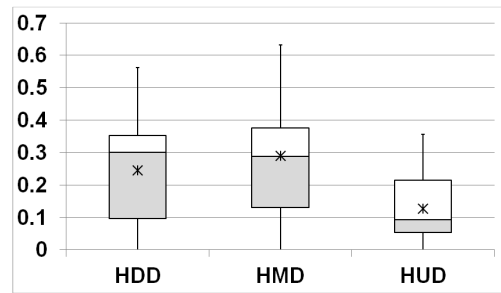


Figure 5. Navigational error ratio (*: mean).

There was a statistically significant difference between the conditions in the number of times the participants exceeded the speed limit ($\chi^2(2)=10.595$, $p=.005$) (see Figure 6). A post hoc analysis found participants exceeded the speed limit significantly more times in the HDD condition (Median [IQR]: 0.5 [0-4.25]) compared the HMD (0 [0-1], $Z=-2.442$, $p=.015$) and HUD (0 [0-1], $Z=-2.675$, $p=.007$), but there was no significant difference between HMD and HUD ($Z=-0.333$, $p=.739$).

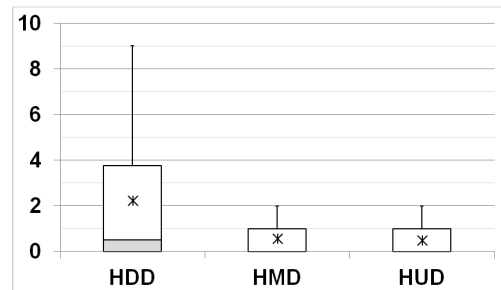


Figure 6. Number of times exceeded speed limit (*: mean).

Figure 7 shows the average amount of time (in seconds) the participants spent above the speed limit. There was a statistically significant difference between the conditions ($\chi^2(2)=7.091$, $p=.029$). A post hoc analysis revealed that the participants exceeded the speed limit significantly

longer in the HDD condition (45 [0-272.75]) compared to the HMD (0 [0-1.75], $Z=-2.432$, $p=.015$) and HUD (0 [0-33.5], $Z=-2.599$, $p=.009$) conditions, while no significant difference was found between the HMD and HUD conditions ($Z=-0.255$, $p=.799$).

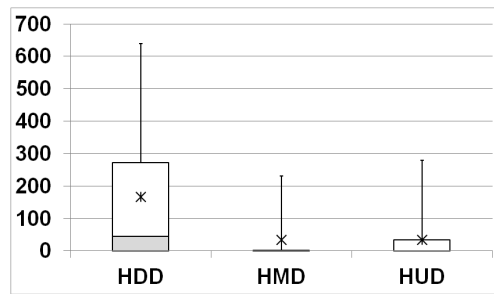


Figure 7. Amount of time participants spent while exceeding the speed limit (in seconds; *: mean).

The results of the number of characters spotted by the participants while driving are summarized in Figure 8. There was a statistically significant difference between the conditions ($\chi^2(2)=6.971$, $p=.031$), and the post hoc analysis showed that there was a statistically significant difference between the HDD (10 [7-12]) and HUD (14 [11.75-18]) conditions in the number of character models found ($Z=-2.953$, $p=.003$). The HMD (12.5 [9.5-7.25]) condition was not significantly different from the other two conditions (HDD: $Z=-2.040$, $p=.041$ and HUD: $Z=-1.357$, $p=.175$).

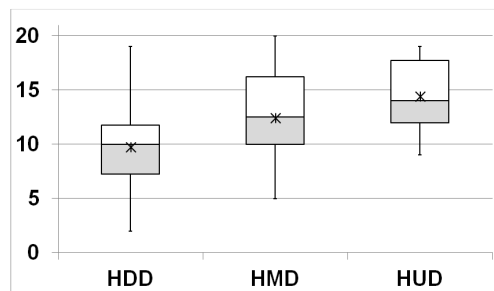


Figure 8. Number of characters spotted (*: mean).

Questionnaire – Rating and Ranking

There were seven Likert scale rating questions on various aspects of usability. Participants answered the question on a scale from 1 (Totally Disagree) to 7 (Totally Agree). Figure 9 summarises the results of the rating questions. To assess the results of each question, we used a

Friedman test ($\alpha=.05$) to evaluate if there were significant difference in rating between the conditions. For post hoc analysis we used Wilcoxon signed-rank tests with Bonferroni correction ($\alpha=.0167$).

The results showed there was no significant difference between the conditions ($\chi^2(2)=4.429$, $p=.109$) in terms of visibility of the surrounding environment (Q1). However, there was a significant difference between the conditions ($\chi^2(2)=18.542$, $p<.001$) in visibility of the displays (Q2). Post hoc analysis with Bonferroni corrected Wilcoxon signed-rank tests revealed the HUD was significantly more visible than the other displays (HDD: $Z=-3.221$, $p=.001$; HMD: $Z=-3.375$, $p=.001$). No significant difference was found between the HDD and HMD ($Z=-.108$, $p=.914$).

For the question asking how distracting the displays are (Q3), participants rated each display significantly differently ($\chi^2(2)=16.097$, $p<.001$). Post hoc analysis revealed that the HUD was significantly less distracting than the HDD ($Z=-2.954$, $p=.003$) or HMD ($Z=-3.363$, $p=.001$), while no significant difference was found between the HDD and HMD ($Z=-1.390$, $p=.165$).

The type of displays also significantly affected the participants perception of ease of navigation (Q4: $\chi^2(2)=15.700$, $p<.001$). Participants felt it was easiest to navigate with the HUD compared to the other two displays (HDD: $Z=-3.037$, $p=.002$; HMD: $Z=-3.347$, $p=.002$). No significant difference was perceived between HDD and HMD ($Z=-.343$, $p=.731$). The HUD was rated as the best by the participants for helping them to multitask (Q5: $\chi^2(2)=12.125$, $p=.002$; vs. HDD: $Z=-2.890$, $p=.004$; vs. HMD: $Z=-3.139$, $p=.002$), while no significant difference was found between the HDD and HMD ($Z=-.779$, $p=.436$).

When asked how helpful was the interface display for navigating properly (Q6: $\chi^2(2)=11.586$, $p=.003$), participants gave significantly better rating to the HUD compared to the HMD ($Z=-3.220$, $p=.001$). However, there was no significant difference between the HUD and HDD conditions ($Z=-2.244$, $p=.025$) nor between the HDD and HMD ($Z=-1.664$, $p=.096$).

Finally, when participants were asked to rate how comfortable they felt using each display for driving (Q7) they rated the displays significantly differently

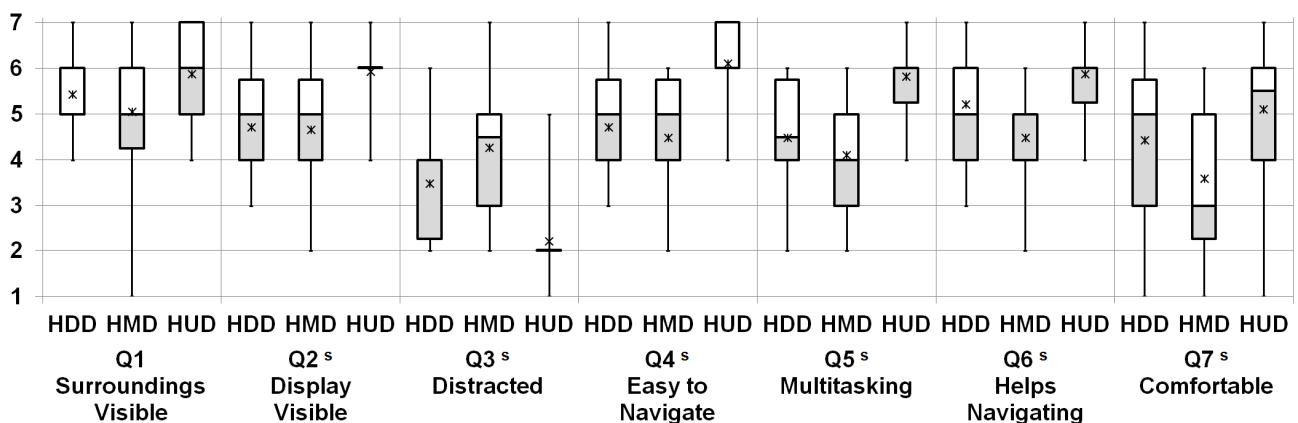


Figure 9. Likert scale rating results (1: Totally disagree ~ 7: Totally agree, *: mean, s: significant difference)

($\chi^2(2)=8.633$, $p=.013$). Post hoc analysis revealed that participants were feeling significantly more comfortable with using HUD compared to HMD ($Z=-2.495$, $p=.013$), yet there was no significant difference between the HDD and HMD ($Z=-1.535$, $p=.125$), nor between the HDD and HUD conditions ($Z=-1.551$, $p=.121$).

After trying all three conditions, participants were asked to rank the displays in the order of their preference, from 1 (the best) to 3 (the worst). Participants ranked the HUD display as the best (*Median [IQR] = 1 [1-1.75]*), while the HDD took the second place (2 [2-2.75]), and the HMD was ranked as the worst (3 [2-3]). Friedman test found the displays were ranked significantly different ($\chi^2(2)=12.366$, $p=.002$). A post hoc analysis with Bonferroni corrected Wilcoxon signed-rank tests revealed that participants ranked the HUD as the most preferable display over the other two display (HMD: $Z=-2.982$, $p=.003$ and HDD: $Z=-2.500$, $p=.012$). However there was no significant difference in participants' preference between HDD and HMD ($Z=-1.380$, $p=.012$).

Questionnaire – Qualitative Feedback

In order to understand more details of the participant's subjective opinion on each display, we asked participants to write comments on what they liked and disliked about each condition. We also asked for any suggestions to modify or add to the displays.

Participants mentioned visibility as the main advantage of HUD. They wrote it was *"simple and easily visible,"* and *"in the line of sight,"* making it *"easy to see the real world."* Some participants saw the display fixed in the space as a positive feature, saying, *"It doesn't move with my line of sight, I can chose to look away when I wish."* But the same feature was also seen as a disadvantage to others as they were not able to see the display when they had to look around for the virtual characters. A participant wrote, *"it is a little hard to see the surroundings."*

For HMD, participants mentioned the display following the view as the main advantage. They wrote that it was *"easy to see even when ... not looking forward,"* hence it was *"easy to notice warning messages,"* and not requiring to *"look back at the road to notice directions."* The main problem of the HMD was mainly about its position. Participants wrote, *"the position of the display is annoying,"* and it is *"difficult to focus on the display and the real world at the same time."* Participants also mentioned that the screen-registered way of showing directions is confusing when looking around. A participant wrote, *"the display moving around is disorienting."*

The HDD was perceived as *"familiar to use,"* and not blocking the user's view. Participants wrote that it is *"fixed and doesn't follow my head,"* and *"out of the way and easier to see the surroundings."* On the other hand, not being within the primary task space was mentioned as the main disadvantage. They wrote, *"(it) required me to look down and take eyes off the road,"* and hence *"I don't notice the change in speed always."*

When asked for the reasoning behind the ratings, the main reason for the HUD being rated the highest was the display being in the field of view while driving so that the participant did not have to move their eye and focus away from the road. As the HDD was rated most commonly in second place, the justification for being lower than the HUD was mainly because it requires the participant to look off the road. For the HMD being rated the lowest, the main reason was the position of the display at the top right of the user's view forced the user to focus on and off each time they had look at it and glance back on the road.

DISCUSSION

Overall, the results indicate HUD as the best in most area, yet each display having their own strength and weak points. In terms of navigational errors, participants performed better while using HUD compared to the other two displays. This may be due to the fact that the windshield was always in the clear field of view of the user while they were looking through it and there is no requirement to glance away from the road (as in the HDD) or to change focus (as in the HMD).

It is also interesting to note that there was a statistically significant reduction in the number of errors when using the HDD over the HMD, despite the HDD requiring the user to look down and away from the road. This may be due to having the display following the user head orientation might cause them to get confused since the instructions shown on the display were relative to the car, not the display. This reasoning is supported by some of the qualitative feedback given where the users complained about the moving display being annoying and confusing.

Despite being statistically better than the HMD at avoiding navigational errors, the HDD falls short in both the number of times it exceeded the speed limit and the time spent above the speed limit. This is probably because the driver has to glance off the road and look at the console to notice changes. Users also mentioned how hard it was to notice warnings on the HDD, which was also identified as an advantage for the other two displays.

In order to test how much the display forces the user to take his or her eyes off the road, virtual characters were added at places along the map. The HUD clearly wins over the HDD in this case, since with the HDD the user is busy with his or her eyes off the road and does not notice the models in the surroundings.

There are limitations with the current study that could be addressed in future research. For the HMD, the navigational instructions were not provided based on relative position of the head orientation, making it slightly more difficult to understand them. The size of the display areas for each AR Display condition was controlled to be the same, which could impact the results for the HUD and HMD where a much larger display area could be used in future setups. The interface layout for each display was the same. Changing the layout of the interface based on the display may affect the outcome of the results. A common layout was used in order to prevent bias in the results based on the layout.

While conducting the study in VR environment gave more control of the conditions, it had limitations of not being able to fully replicate the real world environment (e.g. sun glare, weather effect). Future studies using display setups on real vehicles would need to focus on further investigating such environmental characteristics.

Overall, the main implication of the study can be summarised as the HUD displays are the most desirable type of display for car navigation compared to HDD or HMD given similar size of field of view.

CONCLUSION

In this paper we have presented comparative user evaluations of AR HUD, HMD and HDD navigation aids in a driving simulator. The HUD was found to produce better driving performance, less errors, and was preferred over the HDD. The HMD matched the HUD in enabling the user to monitor their speed, but users thought it was difficult to use and the worst of the display options.

Much of the criticism for the HMD in this study is based on using a monocular peripheral display. With the recent advancements in the field of HMDs, there are many other display options that would be good to explore. In future, it would be good to study how a stereo HMD would compare against a HUD or HDD. Another possibility for future study would be to explore if the HMD would provide navigational instructions relative to the orientation of the head. There are other potential HMD interface configurations that could be tested as well. Finally, the user study was only conducted in a driving simulation. It would be good to test future interfaces in a real car driving outdoors to see how closely the results match those from the virtual simulation.

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